

Clayton Yu

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Education

School of Computer Science – Carnegie Mellon University, Pittsburgh, PA **May 2029**

Bachelor of Science in Computational Biology

GPA: 4.0 / 4.0 – *Dean's List with High Honors*

Current Coursework: Biochemistry I, Great Ideas in Comp. Bio., Quantitative Cell & Molecular Biology Lab, Vector Calculus

Completed Coursework: Principles of Imperative Computation (A), Mathematical Foundations for Computer Science (A)

New Hyde Park Memorial High School, New Hyde Park, NY

Jun 2025

Salutatorian of the Class of 2025

GPA: 102.66 / 100.0 – *High Honor Roll*

Selected AP Courses: Research (5), Biology (5), Chemistry (5), Computer Sci. A (5), Calculus BC (5), Statistics (5)

Research and Laboratory Experience

Albert Einstein College of Medicine, Bronx, NY

Jul 2025 – Aug 2025

Volunteer – Molecular Cytogenetics Core Lab (Dr. Jidong Shan)

- Learned to perform: DNA/RNA Extraction, Running agarose gels, Giemsa staining, micro-pipetting.
- Assisted in Dr. Shan's spectral karyotyping workflow, gaining greater theoretical and practical understanding of karyotyping techniques (SKY, G-banding).

Science Research Program, New Hyde Park Memorial HS, New Hyde Park, NY

Sep 2022 – May 2025

Student Researcher – 2024-25 Individual Research Project

- Investigated the role of autoimmunity in the pathogenesis of Narcolepsy Type I, produced a final paper.
- Analyzed ~260 bulk RNA-seq samples of T cells using differential expression and pathway enrichment analysis.
- Wrote R scripts to access publicly accessible data, preprocess dataset, conduct analysis, and produce graphs.

Hofstra University, Hempstead, NY

Jul 2024 – Aug 2024

Student Researcher – Department of Chemistry (Dr. Daniel P Miller)

- Researched the host-guest binding of organic chelators to Fe(3+) ions using computational chemistry (Gaussian).

AP Research Course, New Hyde Park Memorial HS, New Hyde Park, NY

Sep 2023 – Jun 2024

AP Capstone Student – Individual Research Project

- Applied sentiment analysis to Reddit posts, examining how emoji usage is correlated with user sentiment.
- Found evidence supporting a bimodal distribution in the sentiment distribution of individual emoji characters.

Regeneron Pharmaceuticals, Tarrytown, NY

Jul 2023 – Aug 2023

High School Mentorship Student – Molecular Profiling and Data Science (Alex Zhao, Jinrui Liu)

- Researched the impact of immune stimulation on the transcriptome of Clonal Hematopoiesis (CHIP) patients.
- Learned R, writing scripts to conduct differential expression analysis on RNA-seq data.

Extracurricular Activities

New Hyde Park Science Olympiad Team

Sep 2021 – Jun 2025

President (2024-25), Vice President (2023-24)

- Led 50+ member team in the Western Long Island Regional Science Olympiad.
- Delegated officer roles, organized meetings & merchandise to promote students' interest in science.

New Hyde Park Pep Band

Sep 2023 – Jun 2025

Drum Major/President (2024-25)

- Conducted the New Hyde Park Pep Band in the Fall football season and assisted running classroom rehearsals.
- Led the New Hyde Park Marching Band (80+ members) in Homecoming and Memorial Parades.

Peer Tutoring Center

Oct 2022 – Jun 2025

Volunteer Student Tutor

- Spent 150+ hours instructing 7th - 12th grade students in mathematics (calculus, precalculus, geometry, algebra) and sciences (chemistry, biology, earth science). Adapted learning plans for individual students.

Skills and Awards

Programming: R, Python, C, Java, Shell scripting

Tools: VS Code, Microsoft Office, PyCharm, Overleaf

Lab Skills: Pipetting, Running gels, DNA/RNA Extraction

Musical Interests: A Capella, Piano, French Horn

Selected Awards – Long Island Science Congress Highest Honors, Mel Cowgill Award, Dr. Mofenson Memorial Award; National Merit Finalist; NYSSEF 3rd Place in Life Sciences; 2-time US Chemistry Olympiad National Exam Qualifier